

Create and Run ML Pipelines with Vertex Pipelines

Vertex AI Pipelines (formerly Kubeflow Pipelines on GCP) orchestrate scalable ML workflows using Python SDK. Lab builds pipelines processing text input through 3-step intro workflow and full AutoML classification lifecycle.

- **Core Objectives**
 - Use Kubeflow Pipelines SDK (`kfp.dsl.pipeline``) to define reusable components and orchestrate steps.
 - Leverage `google_cloud_pipeline_components`` library for native Vertex AI integration (training, evaluation, deployment).
- **Pipeline Types**
 - Intro Pipeline: 3-step text processing workflow demonstrating SDK basics and component chaining.
 - AutoML Pipeline: Trains classification model, evaluates performance, deploys endpoint automatically.
- **Advanced Features**
 - Pre-built components handle Vertex AI services without custom containers.
 - Cloud Scheduler automates recurring pipeline executions on cron schedule.

Note: - Task-5 options are deprecated in the GCP lab, so perform the tasks as shown in the images carefully.
- Do not leave any Tasks even though the score count is not available, at the end of the lab the marks are credited after completing the tasks.

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Lab setup instructions and requirements

End Lab

05:14:32

⌚ Time limit

⚠️ Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-02-9a4bf3189b9e

📄

Password

63du0wcZQmsg

📄

GCP Project ID

qwiklabs-gcp-02-48e62cd

📄

Activate Cloud Shell

Cloud Shell is a virtual machine that contains development tools. It offers a persistent 5-GB home directory and runs on Google Cloud. Cloud Shell provides command-line access to your Google Cloud resources. `gcloud` is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab completion.

1. Click the **Activate Cloud Shell** button (🖱️) at the top right of the console.

2. Click **Continue**.

It takes a few moments to provision and connect to the environment. When you are connected, you are also authenticated, and the project is set to your `PROJECT_ID`.

Sample commands

- List the active account name:

gcloud auth list

📄

(Output)

Credentialed accounts:

Lab instructions and tasks

Overview

Setup and requirements

Task 1. Cloud environment setup

Task 2. Create a Cloud Storage bucket

Task 3. Enable the Recommended APIs

Task 4. Launch a Vertex AI Notebooks instance

Task 5. Clone a course repo within your Vertex AI Notebooks instance

Task 6. Create and Run ML Pipelines with Vertex Pipelines

End your lab

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Lab setup instructions and requirements

End Lab

05:13:56

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Open Google Cloud console

Username

student-02-9a4bf3189b9e

Password

63du0wcZQmsg

GCP Project ID

qwiklabs-gcp-02-48e62cd

gcloud auth list

(Output)

Credentialed accounts:

- <myaccount>@<mydomain>.com (active)

(Example output)

Credentialed accounts:

- google1623327_student@qwiklabs.net

List the project ID:

gcloud config list project

(Output)

[core]

Lab instructions and tasks

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https://console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-02-48e62cddb96&pli=1&cloudshell=true

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Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)

Don't show again Dismiss

Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.

Your Cloud Platform project in this session is set to **qwiklabs-gcp-02-48e62cddb96**.

Use ``gcloud config set project [PROJECT_ID]`` to change to a different project.

student_02_9a4bf3189b9e@cloudshell:~ (**qwiklabs-gcp-02-48e62cddb96**)\$ gcloud auth list

Credentialed Accounts

ACTIVE: *

ACCOUNT: student-02-9a4bf3189b9e@qwiklabs.net

To set the active account, run:

\$ gcloud config set account `ACCOUNT`

student_02_9a4bf3189b9e@cloudshell:~ (**qwiklabs-gcp-02-48e62cddb96**)\$ gcloud config list project

[core]

project = qwiklabs-gcp-02-48e62cddb96

Your active configuration is: [cloudshell-4770]

student_02_9a4bf3189b9e@cloudshell:~ (**qwiklabs-gcp-02-48e62cddb96**)\$

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End Lab05:10:15⌚ Time limit

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Open Google Cloud console

Usernamestudent-02-9a4bf3189b9e📄

Password63du0wcZQmsg📄

GCP Project IDqwiklabs-gcp-02-48e62cdl📄

Task 1. Cloud environment setup

Cloud Shell has a few environment variables, including GOOGLE_CLOUD_PROJECT which contains the name of our current Cloud project. We use this in various places throughout this lab. You can see it by running:

echo \$GOOGLE_CLOUD_PROJECT📄

Enable APIs

- In later steps you see where these services are needed (and why), but to begin, run this command to give your project access to the Compute Engine, Container Registry, and Vertex AI services:

gcloud services enable compute.googleapis.com \containerregistry.googleapis.com \aiplatform.googleapis.com \cloudbuild.googleapis.com \cloudfunctions.googleapis.com📄

This should produce a successful message similar to this one:

Lab instructions and tasks

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End your lab

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Open Google Cloud console

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Password63du0wcZQmsg📋

GCP Project IDqwiklabs-gcp-02-48e62cd1📋

Enable APIs

- In later steps you see where these services are needed (and why), but to begin, run this command to give your project access to the Compute Engine, Container Registry, and Vertex AI services:

```
gcloud services enable compute.googleapis.com \
containerregistry.googleapis.com \
aiplatform.googleapis.com \
cloudbuild.googleapis.com \
cloudfunctions.googleapis.com
```

This should produce a successful message similar to this one:

Operation "operations/acf.cc11852d-40af-47ad-9d59-477a12847c9e" finished successfully.

Task 2. Create a Cloud Storage bucket

Lab instructions and tasks

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Task 6. Create and Run ML Pipelines with Vertex Pipelines

End your lab

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```
student_02_9a4bf3189b9e@cloudshell:~ (qwiklabs-gcp-02-48e62cdbdd96) $ echo $GOOGLE_CLOUD_PROJECT
qwiklabs-gcp-02-48e62cdbdd96
student_02_9a4bf3189b9e@cloudshell:~ (qwiklabs-gcp-02-48e62cdbdd96) $ gcloud services enable compute.googleapis.com
containerregistry.googleapis.com \
aiplatform.googleapis.com \
cloudbuild.googleapis.com \
cloudfunctions.googleapis.com
Operation "operations/acf.p2-1004761692446-e0628128-ba0d-4385-b7a6-51b58982825c" finished successfully.
student_02_9a4bf3189b9e@cloudshell:~ (qwiklabs-gcp-02-48e62cdbdd96) $
```


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Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)

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```
student_02_9a4bf3189b9e@cloudshell:~ (qwiklabs-gcp-02-48e62cddb96) $ BUCKET_NAME=gs://$GOOGLE_CLOUD_PROJECT-bucket
gsutil mb -l us-central1 $BUCKET_NAME
Creating gs://qwiklabs-gcp-02-48e62cddb96-bucket/...
student_02_9a4bf3189b9e@cloudshell:~ (qwiklabs-gcp-02-48e62cddb96) $
student_02_9a4bf3189b9e@cloudshell:~ (qwiklabs-gcp-02-48e62cddb96) $ gcloud projects describe $GOOGLE_CLOUD_PROJECT > project-info.txt
PROJECT_NUM=$(cat project-info.txt | sed -nre 's:.+.*projectNumber\: (.*)\:.*$:\1:p')
SVC_ACCOUNT="${PROJECT_NUM//\ '/'}-compute@developer.gserviceaccount.com"
gcloud projects add-iam-policy-binding $GOOGLE_CLOUD_PROJECT --member serviceAccount:$SVC_ACCOUNT --role roles/storage.objectAdmin
Updated IAM policy for project [qwiklabs-gcp-02-48e62cddb96].
bindings:
- members:
  - user:student-02-9a4bf3189b9e@qwiklabs.net
    role: roles/appengine.appAdmin
- members:
  - serviceAccount:qwiklabs-gcp-02-48e62cddb96@qwiklabs-gcp-02-48e62cddb96.iam.gserviceaccount.com
  - user:student-02-9a4bf3189b9e@qwiklabs.net
    role: roles/bigquery.admin
- members:
  - serviceAccount:1004761692446@cloudbuild.gserviceaccount.com
    role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-1004761692446@gcp-sa-cloudbuild.iam.gserviceaccount.com
    role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:service-1004761692446@gcf-admin-robot.iam.gserviceaccount.com
    role: roles/cloudfunctions.serviceAgent
- members:
  - user:student-02-9a4bf3189b9e@qwiklabs.net
    role: roles/compute.admin
```

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📘 Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)

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```
- user:student-02-9a4bf3189b9e@qwiklabs.net
role: roles/iam.serviceAccountAdmin
- members:
- user:student-02-9a4bf3189b9e@qwiklabs.net
role: roles/ml.admin
- members:
- serviceAccount:admiral@qwiklabs-services-prod.iam.gserviceaccount.com
- serviceAccount:qwiklabs-gcp-02-48e62cddb96@qwiklabs-gcp-02-48e62cddb96.iam.gserviceaccount.com
- user:student-02-9a4bf3189b9e@qwiklabs.net
role: roles/owner
- members:
- user:student-02-9a4bf3189b9e@qwiklabs.net
role: roles/serviceusage.serviceUsageAdmin
- members:
- user:student-02-9a4bf3189b9e@qwiklabs.net
role: roles/source.admin
- members:
- serviceAccount:qwiklabs-gcp-02-48e62cddb96@qwiklabs-gcp-02-48e62cddb96.iam.gserviceaccount.com
- user:student-02-9a4bf3189b9e@qwiklabs.net
role: roles/storage.admin
- members:
- serviceAccount:1004761692446-compute@developer.gserviceaccount.com
role: roles/storage.objectAdmin
- members:
- user:student-02-9a4bf3189b9e@qwiklabs.net
role: roles/viewer
etag: BwZGIoejfUo=
version: 1
student_02_9a4bf3189b9e@cloudshell:~ (qwiklabs-gcp-02-48e62cddb96) $
```

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End Lab05:07:17⌚ Time limit

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Open Google Cloud console

Username

student-02-9a4bf3189b9e1📋

Password

63du0wcZQmsg📋

GCP Project ID

qwiklabs-gcp-02-48e62cd1📋

```
member serviceAccount:$SVC_ACCOUNT --role roles/storage.objectAdmin
```

This ensures that Vertex Pipelines has the necessary permissions to write files to this bucket.

Task 3. Enable the Recommended APIs

- In the Google Cloud console, in the **Navigation menu** (☰), click **Vertex AI > Dashboard**.
- Click **Enable All Recommended API**.

Task 4. Launch a Vertex AI Notebooks instance

- In the Google Cloud Console, on the **Navigation Menu**, click **Vertex AI > Workbench**.
- On the User-Managed Notebooks page, click **CREATE NEW**, select **TensorFlow Enterprise 2.11 (Intel® MKL-DNN/MKL)**.

Lab instructions and tasks

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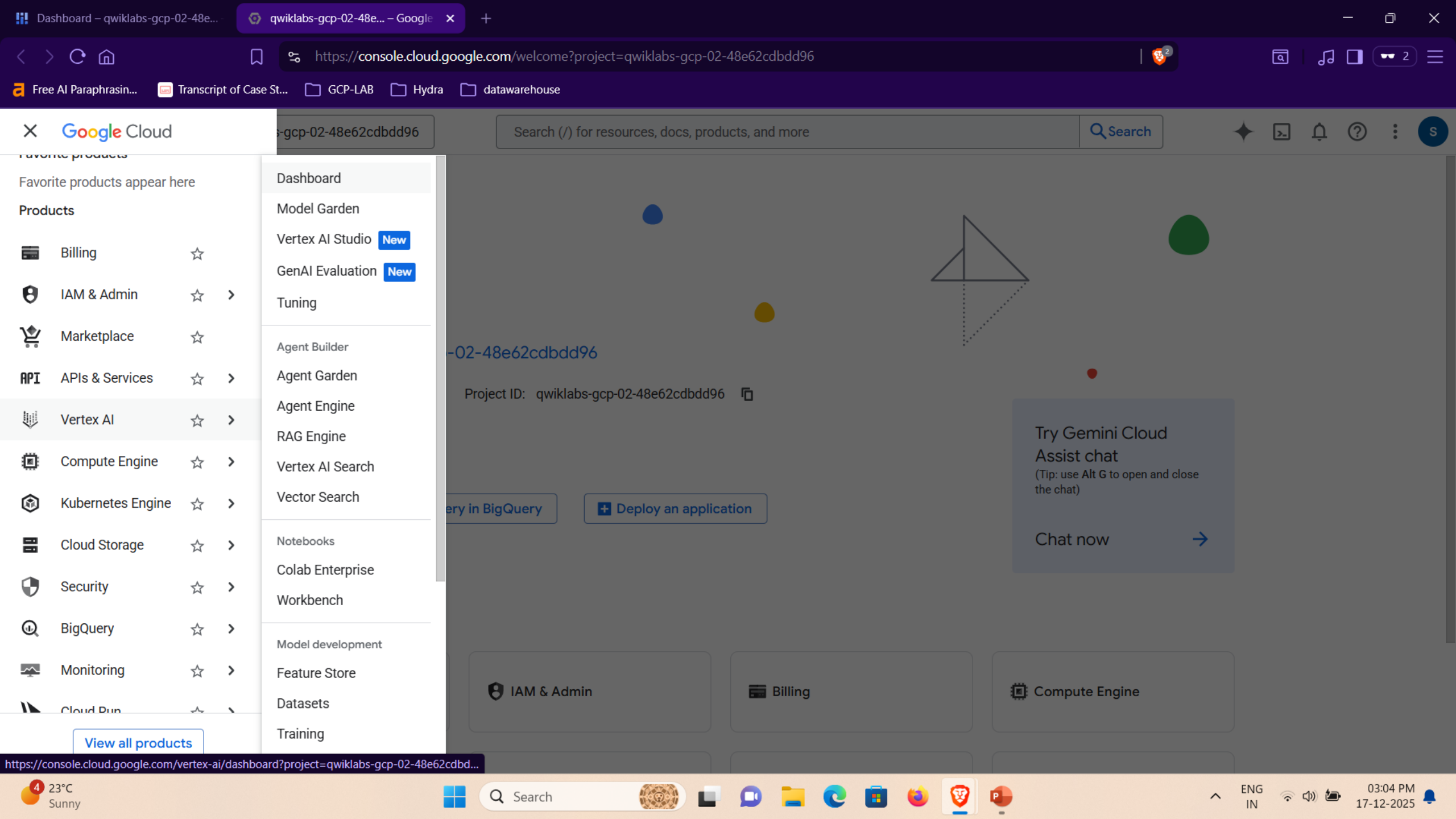
Task 6. Create and Run ML Pipelines with Vertex Pipelines

End your lab

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Get started with Vertex AI

Vertex AI empowers machine learning developers, data scientists, and data engineers to take their projects from ideation to deployment, quickly and cost-effectively. [Learn more about Vertex AI](#) ➔

Enable all recommended APIs

📄 Show API list







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Enable all recommended APIs

▼ Show API list

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✔ Enable service: visionai.googleapis.com Just now
qwiklabs-gcp-02-48e62cdbdd96

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cloudapiregistry.googleapis.com
qwiklabs-gcp-02-48e62cddb96

✔ Enable service: telemetry.googleapis.com Just now
qwiklabs-gcp-02-48e62cddb96

✔ Enable service: dataform.googleapis.com Just now
qwiklabs-qcp-02-48e62cddb96

✔ Enable service: [datalineage.googleapis.com](https://data.lineage.googleapis.com) Just now
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✔ Enable service: notebooks.googleapis.com Just now
qwiklabs-qcp-02-48e62cddb96

✔ Enable service: dataplex.googleapis.com Just now
qwiklabs-gcp-02-48e62cdbdd96

✔ Enable service: compute.googleapis.com Just now
qwiklabs-qcp-02-48e62cddb96

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artifactregistry.googleapis.com

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Get started with Vertex AI

Vertex AI empowers machine learning developers, data scientists, and data engineers to take their projects from ideation to deployment, quickly and cost-effectively. [Learn more about Vertex AI](#)

Enable all recommended APIs

📉 Show API list

✔️ Enable service: datalineage.googleapis.com Just now
qwiklabs-gcp-02-48e62cdbdd96

✔️ Enable service: notebooks.googleapis.com Just now
qwiklabs-gcp-02-48e62cdbdd96

✔️ Enable service: dataplex.googleapis.com Just now
qwiklabs-gcp-02-48e62cdbdd96

✔️ Enable service: compute.googleapis.com Just now
qwiklabs-gcp-02-48e62cdbdd96

✔️ Enable service: artifactregistry.googleapis.com Just now
qwiklabs-gcp-02-48e62cdbdd96

✔️ Enable service: dataflow.googleapis.com Just now
qwiklabs-gcp-02-48e62cdbdd96

✔️ Enable service: storage-component.googleapis.com Just now
qwiklabs-gcp-02-48e62cdbdd96

✔️ Enable service: aiplatform.googleapis.com Just now
qwiklabs-gcp-02-48e62cdbdd96

See all activities

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DC Infotech Internship Portal

Create and Run ML Pipelines with Vertex AI

https://partner.skills.google/focuses/41026?catalog_rank=%7B"rank"%3A1%2C"num_filters"%3A0%2C"has_search"%3Atrue%7D"rank"%3A1%2C"num_filters"%3A0%2C"has_search"%3Atrue%7D

Free AI Paraphrasing Tool

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GCP-LAB

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datawarehouse

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Partner

What do you want to learn today?

6058

1

Create and Run ML Pipelines with Vertex Pipelines

Lab setup instructions and requirements

End Lab

05:04:32

Time limit

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-02-9a4bf3189b9e0

Password

63du0wcZQmsg

GCP Project ID

qwiklabs-gcp-02-48e62cd1

Task 4. Launch a Vertex AI Notebooks instance

1. In the Google Cloud Console, on the **Navigation Menu**, click **Vertex AI** > **Workbench**.

2. On the User-Managed Notebooks page, click **CREATE NEW**, select **TensorFlow Enterprise 2.11 (Intel® MKL-DNN/MKL)**.

3. In the **New notebook** instance dialog, confirm the name of the deep learning VM, if you don't want to change the region and zone, leave all settings as they are and then click **Create**. The new VM will take 2-3 minutes to start.

4. Click **Open JupyterLab**.
A JupyterLab window will open in a new tab.

5. If you see "Build recommended" pop up, click **Build**. If you see the build failed, ignore it.

Task 5. Clone a course repo within your Vertex AI Notebooks instance

Lab instructions and tasks

Overview

Setup and requirements

Task 1. Cloud environment setup

Task 2. Create a Cloud Storage bucket

Task 3. Enable the Recommended APIs

Task 4. Launch a Vertex AI Notebooks instance

Task 5. Clone a course repo within your Vertex AI Notebooks instance

Task 6. Create and Run ML Pipelines with Vertex Pipelines

End your lab

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Dashboard – qwiklabs-gcp-02-48e...

Dashboard – Vertex AI – qwiklab...

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🔍 Free AI Paraphrasin... 📄 Transcript of Case St... 📁 GCP-LAB 📁 Hydra 📁 datawarehouse

☰ Google Cloud

🔍 qwiklabs-gcp-02-48e62cdbdd96

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🏠 Vertex AI ☆ Dashboard

Agent Builder ^
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🔗 Vector Search
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Enable all recommended APIs

Tutorials

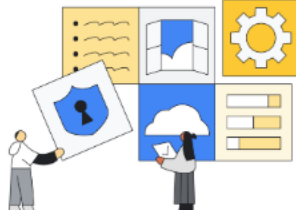
Learn how to use generative AI, Gemini, and more on Vertex AI.

View tutorials

▼ Show API list







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Dashboard – qwiklabs-gcp-02-48e...

Workbench – Vertex AI – qwiklab

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🔗 https://console.cloud.google.com/vertex-ai/workbench/user-managed?project=qwiklabs-gcp-02-48e62cdbdd96

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🔗 Free AI Paraphrasin...

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🏠 Vertex AI / Workbench / User-managed notebooks

Agent Builder

🤖 Agent Garden

🤖 Agent Engine

📄 RAG Engine

🔍 Vertex AI Search

🔗 Vector Search

Notebooks

📄 Colab Enterprise

📄 Workbench

Model development

📄 Feature Store

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🔗 Training

📄 Experiments

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🔗 Pipelines

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Workbench

+ Create New

🔄 Refresh

📖 Learn

Instances

Executions

Schedules

View:

Instances

User-managed Notebooks

Managed Notebooks

⚠️ Vertex AI Workbench User-managed Notebooks is deprecated and support will end soon. We recommend that you migrate to Vertex AI Workbench instances.

Migrate

Notebooks have JupyterLab 3 pre-installed and are configured with GPU-enabled machine learning frameworks. [Learn more](#)

🔍 Filter Enter property name or value

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☐	☒	Notebook name ↑	Zone	Auto upgrade	Environment	Machine Type	GPUs	Owner	Created
--------------------------	----------------------------------	-----------------	------	--------------	-------------	--------------	------	-------	---------

Note:

• During this lab User-managed Notebook was deprecated, hence ‘Create New’ option was not working

• Follow the below steps shown in the images very carefully, and if some options do not match as shown in the images:

• Refresh the tab ,or

• Start the lab again.

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Dashboard – qwiklabs-gcp-02-48e...

Workbench – Vertex AI – qwiklab

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Workbench

+ Create New

🔄 Refresh

Instances Executions Schedules

View: Instances User-managed Notebooks Managed Notebooks

📘 JupyterLab 4 is now generally available in Vertex AI Workbench.

Workbench Instances have JupyterLab 4 pre-installed and are configured with GPU-enabled machine learning frameworks. [Learn more](#)

Filter

Enter property name or value

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☐	☒	Instance name ↑	Zone	Auto upgrade	Version	Machine Type	GPUs	Owner	Created	Labels
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Note:

- During this lab User-managed Notebook was deprecated.
- Instead ‘Instances’ was used to perform the lab.
- Follow the below steps shown in the images very carefully, and if some options do not match as shown in the images:
 - Refresh the tab ,or
 - Start the lab again.

Dismiss



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Workbench – Vertex AI – qwiklab

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Workbench

+ Create New

↻ Refresh

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Managed Notebooks

📘 JupyterLab 4 is now generally available in Vertex AI Workbench.

Workbench Instances have JupyterLab 4 pre-installed and are configured with GPU-enabled machine learning frameworks. [Learn more](#)

Filter

Enter property name or value

☐ ●

Instance name ↑

Zone

Auto upgrade

Version

New instance

Name *

instance-20251217-150931

Must start with a letter followed by up to 47 lowercase letters, numbers, or hyphens (-) and cannot end with a hyphen

Region *

us-central1 (Iowa)

Zone *

us-central1-a

Some regions are restricted due to a policy set by your organization. [Learn more](#)

☐ Attach 1 NVIDIA T4 GPU

☒ Enable Apache Spark and BigQuery Kernels

☒ Network in this project

☐ Shared network

Network

default

Subnetwork *

default(10.128.0.0/20)

Instance properties

Machine type

e2-standard-4

Data disk

100 GB Balanced persistent disk

Permission

Compute Engine default service account

JupyterLab version

JupyterLab 4

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Managed Notebooks

📘 JupyterLab 4 is now generally available in Vertex AI Workbench.

Workbench Instances have JupyterLab 4 pre-installed and are configured with GPU-enabled machine learning frameworks. [Learn more](#)

Filter

Enter property name or value

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Instance name ↑

Zone

Auto upgrade

Version

Must start with a letter followed by up to 47 lowercase letters, numbers, or hyphens (-) and cannot end with a hyphen

Region * us-central1 (Iowa) ?

Zone * us-central1-a ?

Some regions are restricted due to a policy set by your organization. [Learn more](#)

☐ Attach 1 NVIDIA T4 GPU

☒ Enable Apache Spark and BigQuery Kernels

☒ Network in this project

☐ Shared network

Network default ?

Subnetwork * default(10.128.0.0/20) ?

Instance properties

Machine type

e2-standard-4

Data disk

100 GB Balanced persistent disk

Permission

Compute Engine default service account

JupyterLab version

JupyterLab 4

Estimated cost ?

\$145.20 monthly, \$0.20 hourly

Advanced options

Cancel

Create

4

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Workbench – Vertex AI – qwiklab

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📁 Workbench

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Tutorials

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Workbench

+ Create New

↺ Refresh

Learn

Instances

Executions

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View:

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User-managed Notebooks

Managed Notebooks

📘 JupyterLab 4 is now generally available in Vertex AI Workbench.

Dismiss

Workbench Instances have JupyterLab 4 pre-installed and are configured with GPU-enabled machine learning frameworks. [Learn more](#)

Filter

Enter property name or value

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☐	☒	Instance name ↑	Zone	Auto upgrade	Version	Machine Type	GPUs	Owner	Cr
☐	☒	instance-20251217-151114	us-central1-a	—	M137	Efficient Instance: 4 vCPUs, 16 GB RAM	None	1004761692446-compute@developer.gserviceaccount.com	De 20 3:1

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Dashboard – qwiklabs-gcp-02-48e...

Workbench – Vertex AI – qwiklabs-gc

JupyterLab

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Free AI Paraphrasin... Transcript of Case St... GCP-LAB Hydra datawarehouse

File Edit View Run Kernel Tabs Settings Help

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Name	Modified
📄 notebook_templat...	7m ago
• 📄 Untitled.ipynb	now

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Terminal 1

Launcher

📖 Notebook

🐍

Python 3
(ipykernel)
(Local)

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PyTorch 1-13
(Local)

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TensorFlow 2-11
(Local)

>_ Console

🐍

Python 3
(ipykernel)
(Local)

🔥

PyTorch 1-13
(Local)

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TensorFlow 2-11
(Local)

\$_ Other

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Terminal

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Text File

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Markdown File

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Python File

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Show Contextual
Help

Note:

- JupiterLab may take around 10 to 20 minutes to start.
- Keep refreshing the page or keep opening the link again.

Would you like to get notified about official
Jupyter news?

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Open privacy policy

Yes

No

Simple

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Dashboard – qwiklabs-gcp-02-48e...Workbench – Vertex AI – qwiklabs-gc...lab - JupyterLab

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Free AI Paraphrasin...Transcript of Case St...GCP-LABHydradatawarehouse

FileEditViewRunKernelTabsSettingsHelp

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Name	Modified
training-data-anal...	10s ago
notebook_templat...	10m ago
• Untitled.ipynb	1m ago

Terminal 1

Launcher

```
(base) jupyter@instance-20251217-151114:~$ git clone https://github.com/GoogleCloudPlatform/training-data-analyst
Cloning into 'training-data-analyst'...
remote: Enumerating objects: 68305, done.
remote: Counting objects: 100% (10/10), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 68305 (delta 4), reused 3 (delta 3), pack-reused 68295 (from 3)
Receiving objects: 100% (68305/68305), 706.02 MiB | 29.37 MiB/s, done.
Resolving deltas: 100% (43951/43951), done.
Updating files: 100% (12917/12917), done.
(base) jupyter@instance-20251217-151114:~$
(base) jupyter@instance-20251217-151114:~$ ls
Untitled.ipynb  notebook_template.ipynb  training-data-analyst
(base) jupyter@instance-20251217-151114:~$
(base) jupyter@instance-20251217-151114:~$ cd training-data-analyst
(base) jupyter@instance-20251217-151114:~/training-data-analyst$
(base) jupyter@instance-20251217-151114:~/training-data-analyst$ ls
CODEOWNERS      CPB100  blogs      courses  doc  learning_rate.ipynb  self-paced-labs
CONTRIBUTING.md LICENSE  bootcamps  datalab  iot  quests
(base) jupyter@instance-20251217-151114:~/training-data-analyst$
```

Simple

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Terminal 1

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Lab setup instructions
and requirements

End Lab

04:45:15

② Time limit

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked.
[Learn more.](#)

[Open Google Cloud console](#)

Username

student-02-9a4bf3189b9e0

Password

63du0wcZQmsg

GCP Project ID

qwiklabs-qcp-02-48e62cd1

1. In the notebook interface, navigate to **training-data-analyst > courses > machine_learning > deeplive2 > machine_learning_in_the_enterprise > labs**, and open **pipelines_intro_kfp.ipynb**.
2. In the notebook interface, click **Edit > Clear All Outputs**.
3. Carefully read through the notebook instructions and fill in lines marked with **#TODO** where you need to complete the code.

Tip: To run the current cell, click the cell and press SHIFT+ENTER. Other cell commands are listed in the notebook UI under **Run**.

- Hints may also be provided for the tasks to guide you along. Highlight the text to read the hints (they are in white text).
- If you need more help, look at the complete solution at **training-data-analyst > courses > machine_learning > deepdive2 > machine_learning_in_the_enterprise > solutions**, and open **pipelines intro kfp.ipynb**.

End your lab

Note:

- The lab does not have any score counting appearing.
- Even so do at least minimum of 4 #TODO tasks to complete the task-6 and get full marks once the lab is ended.

Lab instructions and tasks

Overview

Setup and requirements

Task 1. Cloud environment setup

Task 2. Create a Cloud Storage bucket

Task 3. Enable the Recommended APIs

Task 4. Launch a Vertex AI Notebooks instance

Task 5. Clone a course repo within your Vertex AI Notebooks instance

Task 6. Create and Run ML Pipelines with Vertex Pipelines

End your lab

Dashboard – qwiklabs-gcp-02-48e...

Workbench – Vertex AI – qwiklabs-gc...

lab - JupyterLab

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📁 GCP-LAB

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📁 datawarehouse

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Name	Modified
📁 training-data-anal...	now
📄 notebook_templat...	10m ago
• 📄 Untitled.ipynb	1m ago

Terminal 1

Launcher

```
(base) jupyter@instance-20251217-151114:~/training-data-analyst$ cd courses
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses$
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses$ cd machine_learning
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning$
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning$
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning$ cd labs
bash: cd: labs: No such file or directory
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning$
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning$ cd deepdive2 > machine_learning_in_the_enterprise > labs
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning/deepdive2$ cd machine_learning_in_the_enterprise > labs
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning/deepdive2/machine_learning_in_the_enterprise$
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning/deepdive2/machine_learning_in_the_enterprise$ cd labs
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning/deepdive2/machine_learning_in_the_enterprise/labs$
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning/deepdive2/machine_learning_in_the_enterprise/labs$
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning/deepdive2/machine_learning_in_the_enterprise/labs$
```

Simple

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Terminal 1 1 🔔

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Dashboard – qwiklabs-gcp-02-48e...Workbench – Vertex AI – qwiklabs-gc...lab - JupyterLab

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Name	Modified
📁 training-data-anal...	now
📄 notebook_templat...	10m ago
• 📄 Untitled.ipynb	1m ago

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```
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning/deepdive2/machine_learning_in_the_enterprise/labs$ ls
bqml-online-prediction.ipynb
build_model_experimentation_lineage_with_prebuild_code.ipynb
comparing_local_trained_models.ipynb
create_run_vertex_pipeline.ipynb
distributed-hyperparameter-tuning.ipynb
gapic-vizier-multi-objective-optimization.ipynb
intro_to_vertex_pipelines.ipynb
model_monitoring.ipynb
model_monitoring_for_custom_model_batch_prediction_job.ipynb
overfit_and_underfit.ipynb
pipelines_intro_kfp.ipynb
sdk-custom-image-classification-batch.ipynb
sdk_custom_tabular_regression_online_explain.ipynb
sdk_custom_xgboost.ipynb
taxicab_traffic
taxifare
tuning_results.png
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning/deepdive2/machine_learning_in_the_enterprise/labs$
(base) jupyter@instance-20251217-151114:~/training-data-analyst/courses/machine_learning/deepdive2/machine_learning_in_the_enterprise/labs$
```

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• 📄 Untitled.ipynb	1m ago

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GNU nano 5.4

pipelines_intro_kfp.ipynb *

```
"cell_type": "code",
"execution_count": 15,
"metadata": {
  "id": "mbpw7oyM2Ou5"
},
"outputs": [],
"source": [
  "# Initialize the Vertex AI SDK for your project and bucket\n",
  "# TODO 1: from google.cloud import aiplatform

aiplatform.init(
  project=PROJECT_ID,
  location=LOCATION,
  staging_bucket=BUCKET_URI,
)
"

]
},
{
  "cell_type": "markdown",
  "metadata": {
    "id": "define_component:hello_world"
  }
}
```

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Terminal 1

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GNU nano 5.4

pipelines_intro_kfp.ipynb *

```
},
"outputs": [
  {
    "name": "stderr",
    "output_type": "stream",
    "text": [
      "/home/jupyter/.local/lib/python3.7/site-packages/kfp/v2/compiler/compiler.py:1293: FutureWarning: APIs i
      " category=FutureWarning,\n"
    ]
  }
],
"source": [
  "# Compile the pipeline\n",
  "from kfp.v2 import compiler # noqa: F811\n",
  "\n",
  "compiler.Compiler().compile(# TODO 2: compiler.Compiler().compile(
    pipeline_func=pipeline,          # or your pipeline function name
    package_path="pipeline_job.json", # or whatever the notebook specified
  )
]"
],
},
```

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GNU nano 5.4

pipelines_intro_kfp.ipynb *

```
"PipelineJob projects/192756378758/locations/us-central1/pipelineJobs/hello-world-v2-20220922141223 curre>
"PipelineState.PIPELINE_STATE_RUNNING\n",
"PipelineJob projects/192756378758/locations/us-central1/pipelineJobs/hello-world-v2-20220922141223 curre>
"PipelineState.PIPELINE_STATE_RUNNING\n",
"PipelineJob projects/192756378758/locations/us-central1/pipelineJobs/hello-world-v2-20220922141223 curre>
"PipelineState.PIPELINE_STATE_RUNNING\n",
"PipelineJob run completed. Resource name: projects/192756378758/locations/us-central1/pipelineJobs/hello>
]
}
],
"source": [
  "# Create and run the pipeline job run\n",
  "DISPLAY_NAME = \"intro_\" + TIMESTAMP\n",
  "\n",
  "job = # TODO 3: from google.cloud import aiplatform as aip # this is typically already imported earlier

job = aip.PipelineJob(
  display_name=DISPLAY_NAME,
  template_path="pipeline_job.json", # or the same name you used in compile()
  pipeline_root=PIPELINE_ROOT, # usually defined earlier in the notebook
)
\n",
```

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▲ Lab

Migrate to Cloud SQL for PostgreSQL Using Database Migration Service

In this lab, you migrate a stand-alone PostgreSQL database (running on a virtual machine) to Cloud SQL for PostgreSQL using a continuous Databa...

🕒 1 hour

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▲ Lab

Connect an App to a Cloud SQL for PostgreSQL Instance

In this lab, you create a Kubernetes cluster and deploy a simple application to that cluster. Then, you connect the application to the supplied...

🕒 1 hour

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▲ Lab

Securing a Cloud SQL for PostgreSQL Instance

In this lab, you deploy a new Cloud SQL instance using a customer-managed encryption key (CMEK). You configure pgAudit to selectively...

🕒 35 minutes

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▲ Lab

Configure Replication and Enable Point-in-Time Recovery for Cloud SQL

In this lab, you configure replication and point-in-time recovery for a Cloud SQL instance.

🕒 35 minutes

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▲ Lab

Create and Manage Cloud SQL

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Course Badge

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LAB COMPLETED ✓

Create and Run ML Pipelines with Vertex Pipelines

Experience points

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